

```
1  module display (
2      input          clk_i,
3      input          rst_ni,
4      input [2:0]    column_i ,
5      input          row_i,
6      output reg [7:0]  HEX0,
7      output reg [7:0]  HEX1,
8      output reg [7:0]  HEX2,
9      output reg [7:0]  HEX3,
10     output reg [7:0]  HEX4,
11     output reg [7:0]  HEX5
12
13 );
14
15 localparam upper_mask = 8'b1001_1100;
16 localparam lower_mask = 8'b1010_0011;
17
18 always @(posedge clk_i or negedge rst_ni) begin
19     if (!rst_ni) begin
20         HEX0 <= 8'hFF;
21         HEX1 <= 8'hFF;
22         HEX2 <= 8'hFF;
23         HEX3 <= 8'hFF;
24         HEX4 <= 8'hFF;
25         HEX5 <= 8'hFF;
26     end else begin
27
28         // Turn off all displays mask (common anode)
29         HEX0 <= 8'hFF;
30         HEX1 <= 8'hFF;
31         HEX2 <= 8'hFF;
32         HEX3 <= 8'hFF;
33         HEX4 <= 8'hFF;
34         HEX5 <= 8'hFF;
35
36         // Overwrite only selected display
37         case (column_i)
38             3'd0: HEX0 <= row_i ? upper_mask : lower_mask;
39             3'd1: HEX1 <= row_i ? upper_mask : lower_mask;
40             3'd2: HEX2 <= row_i ? upper_mask : lower_mask;
41             3'd3: HEX3 <= row_i ? upper_mask : lower_mask;
42             3'd4: HEX4 <= row_i ? upper_mask : lower_mask;
43             3'd5: HEX5 <= row_i ? upper_mask : lower_mask;
44         endcase
45     end
46 end
47
48 endmodule
```